

PLACEMENT PREDICTOR

Machine Learning se jaano ki aapki placement hogi ya nahi!

Flask (Python) • HTML/CSS/JavaScript • scikit-learn • Random Forest

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🌐 Ye Website Kya Karti Hai?

Idea ek line me: student ka data daalo → ML model prediction dega

Kaise Kaam Karta Hai?

- Student apni details bharta hai (10th, 12th, CGPA, etc.)
- Website ye data Machine Learning model ko bhejti hai
- Model predict karta hai: placement hogi ya nahi
- Saath me chance ka % bhi dikhata hai (gauge me)
- Placed dikha to confetti celebration bhi hoti hai!

Form me 8 Details Bharti Hain

• 10th Percentage (%)	e.g. 85
• 12th Percentage (%)	e.g. 80
• CGPA (0 - 10)	e.g. 8.2
• Internships ki sankhya	e.g. 2
• Projects ki sankhya	e.g. 4
• Aptitude Score (0-100)	e.g. 78
• Communication Score (0-100)	e.g. 82
• Work Experience (Yes/No)	Yes / No

Technology aur Languages

Website 3 hisso se bani hai: Backend + ML + Frontend

Python 3.13

Backend language

Flask app + ML model — pura dimaag isi me hai

Flask

Web Framework

Python ki lightweight library — website chalati hai

scikit-learn

ML Library

Random Forest model train karne ke liye

HTML5

Page Structure

Website ka skeleton — forms, buttons, result box

CSS3

Design + Animation

Dark theme, glassmorphism, neon glow, animations

JavaScript

Interactivity

Form data bhejna, animations, cursor, confetti

Project Structure

Har file ka kaam ek line me

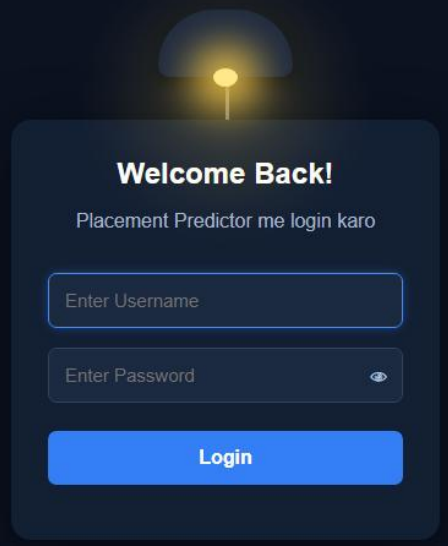
```
placement-predictor/
|
|-- app.py              <- Flask website (backend)
|-- train_model.py      <- ML model train karta hai
|-- model.pkl           <- Trained model (auto save)
|-- placement_data.csv  <- Dataset (2000 students)
|-- requirements.txt    <- zaroori packages
|-- runtime.txt         <- Python version
|
|-- templates/
|   |-- index.html      <- Main website page
|   |-- login.html      <- Login page
|
|-- static/
|   |-- style.css       <- Design + animations
|   |-- script.js       <- Website ka JS logic
|   |-- bg.js           <- (backup background file)
```

Files ka Kaam

- app.py — website ka dil: routes, login, predict API
- train_model.py — dataset + model banata hai
- model.pkl — trained Random Forest (auto load)
- index.html — main page ka structure
- login.html — login page (inline design)
- style.css — poora design + 10+ animations
- script.js — form logic + cursor + scan + confetti

🔒 Login Page

Bina login kiye website nahi khulti — session security



Login Page Features

- Lamp + bulb animation (upar latka hua lamp)
- Username + Password form (POST se bhejta hai)
- Password hide/show ka 👁 toggle button
- Galat password par red error message
- Session server par save hota hai (secret_key)
- Credentials app.py me defined hain

Security: @login_required decorator har page ko protect karta hai — bina login ke / par redirect ho jata hai.

🔧 Login ka Backend Code (app.py)

Flask routes + session + decorator — sab simple hai

```
# Login credentials
VALID_USERNAME = "vikas yadav"
VALID_PASSWORD = "pit jaunpur"

# Har page ko protect karne wala decorator
def login_required(f):
    @wraps(f)
    def wrapper(*args, **kwargs):
        if not session.get("logged_in"):
            return redirect(url_for("login"))
        return f(*args, **kwargs)
    return wrapper

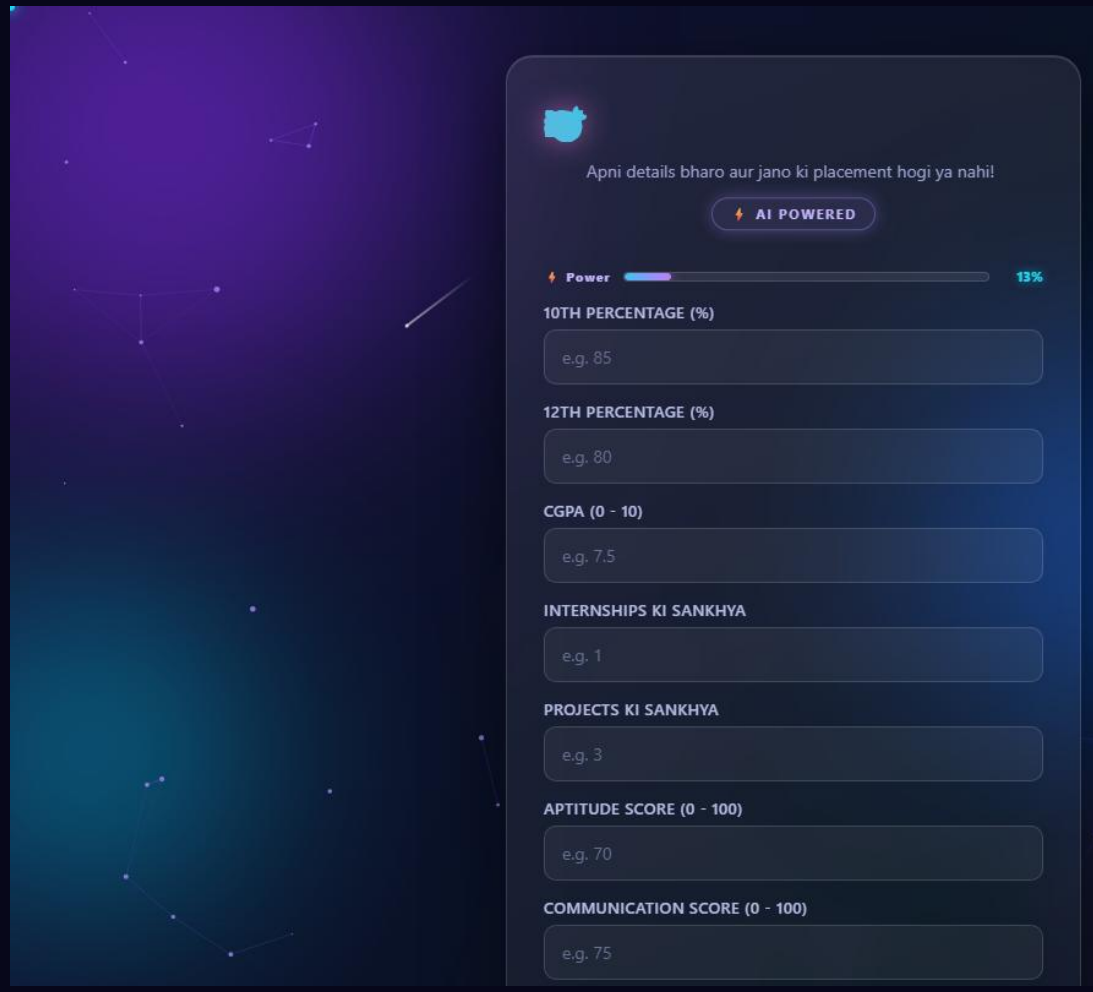
# Login route — GET se form, POST se check
@app.route("/login", methods=["GET", "POST"])
def login():
    error = ""
    if request.method == "POST":
        username = request.form.get("username", "")
        password = request.form.get("password", "")
        if username == VALID_USERNAME \
           and password == VALID_PASSWORD:
            session["logged_in"] = True
            return redirect(url_for("home"))
        error = "Galat username ya password!"
    return render_template("login.html", error=error)
```

Simple Bhasha Me

- session — browser ke liye "identity card"
- session['logged_in'] = True matlab login ho gaya
- Decorator har page par check karta hai
- Galat password par error variable me message
- render_template() HTML file browser ko bhejta hai
- url_for() sahi page ka path dhoondta hai

Main Website Page

Dark theme + glassmorphism card + animated background

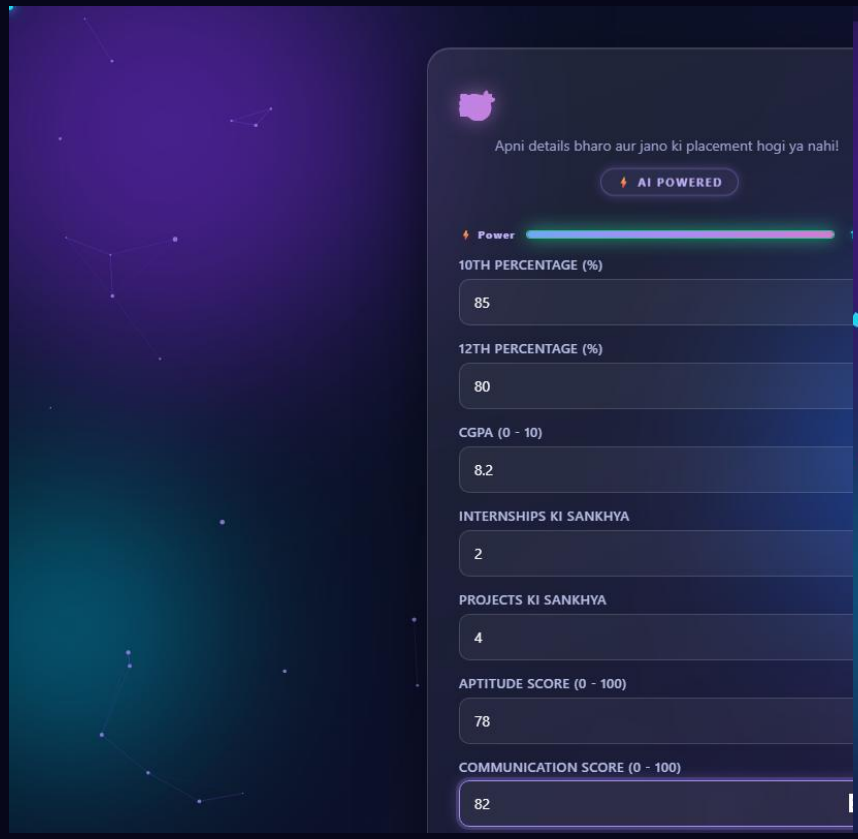


Screen par Kya Dikhta Hai

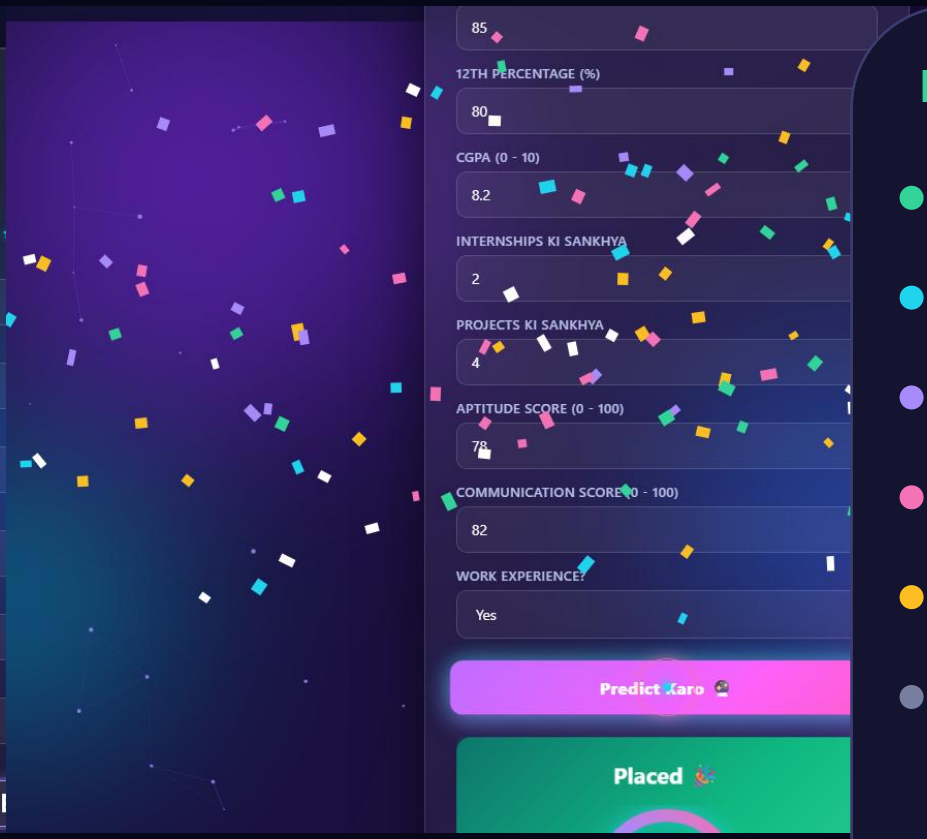
- Animated background — particles + shooting stars
- Floating glowing orbs (mouse ke saath chalti hain)
- Glassmorphism card — mouse par 3D tilt hota hai
- Gradient animated title + typewriter subtitle
- Power bar — fields bharo utna % badhta hai
- Magnetic Predict button (cursor ko khichta hai)
- Neon 'Created by Vikas Yadav' credit

Form Bharo → Prediction Lo

Form bharne par power bar full, submit par AI scan + result



① Form bharte waqt — Power bar badhta hai



② Predict ke baad — Gauge + confetti

Result Me Kya Milta Hai

- Placed / Not Placed
- Chance ka exact %
- Neon circular gauge
- Confetti celebration
- Hologram glitch reveal
- AI scan line animation

Prediction Ka Code (app.py)

Form ka data → model → JSON response

```
# Form se aane wale fields (same order me model ko bhejte hain)
FEATURES = [
    "tenth_percent", "twelfth_percent", "cgpa",
    "internships", "projects", "aptitude_score",
    "communication_score", "work_experience",
]

@app.route("/predict", methods=["POST"])
@login_required
def predict():
    # JavaScript se student ka data aata hai (JSON)
    data = request.get_json()

    # Form ki values ko number me badlo
    values = [float(data[f]) for f in FEATURES]

    # Model ko do — lag 0.5 se placement ka chance
    prob = model.predict_proba([values])[0][1]

    placed = prob >= 0.5
    percent = round(prob * 100, 1)

    # Browser ko JSON wapas bhejo
    return jsonify({
        "prediction": "Placed" if placed else "Not Placed",
        "probability": percent,
        "message": "Strong profile!" if placed else "Skills
improve karo!"
```

Flow Samjho

1. JavaScript fetch() se JSON bhejta hai
2. Flask request.get_json() me padhta hai
3. model.predict_proba() chance nikalta hai
4. probability >= 50 → Placed, warna Not
5. jsonify() se browser ko JSON wapas
6. JS result ko animation ke saath dikhata hai

🧠 Machine Learning Model

Random Forest Classifier — scikit-learn se train hua

```
# train_model.py ka core hissa

# 1. Dataset banaya — 2000 students
n = 2000
tenth = rng.normal(78, 9, n)     # 10th marks
cgpa  = rng.normal(7.5, 0.9, n)  # CGPA
# ... 8 features total

# 2. Placement chance ka formula
chance = (cgpa/10)*30 + (tenth/100)*15
       + internships*8 + projects*2
       + aptitude*0.15 + communication*0.15
       + work_exp*5
placed = (chance + noise > 75).astype(int)

# 3. Model train
model = RandomForestClassifier(n_estimators=100)
model.fit(X_train, y_train)

# 4. Save for website
joblib.dump(model, "model.pkl")
# accuracy: ~87%
```

Random Forest Kya Hai?

- 100 chhote decision trees ka group
- Har tree apna guess karta hai
- Sabka majority vote = final answer
- 8 features input: marks, CGPA, skills...
- Output: 0 (Not Placed) ya 1 (Placed)
- predict_proba() se % chance milta hai
- Dataset: placement_data.csv (2000 rows)

🏠 HTML — Website ka Skeleton

index.html — ye structure browser ko bhejta hai

```
<!-- templates/index.html (simplified) -->

<canvas id="bgParticles"></canvas>  <!-- background -->

<div class="container">             <!-- glass card -->
  <header>
    <h1 id="mainTitle">Placement Predictor</h1>
    <p id="tagline">Apni details bharo...</p>
    <div class="badge">AI Powered</div>
  </header>

  <div class="charge">...</div>     <!-- power bar -->

  <form id="predictForm">
    <label>10th Percentage</label>
    <input type="number" id="tenth_percent" />
    <!-- ... 8 fields total ... -->
    <button type="submit">Predict</button>
  </form>

  <div id="result" class="hidden">  <!-- result box -->
    <h2 id="predictionText"></h2>
    <div class="gauge">             <!-- neon ring -->
      <svg>...</svg>
    </div>
    <p id="messageText"></p>
  </div>
</div>
```

HTML Kya Fix Karta Hai

- canvas — animated background ka drawing area
- form + input — user ki 8 details
- id='result' — JS isi me prediction likhta hai
- SVG circle — neon gauge ring banati hai
- hidden class — result pehle chhupa rehta hai
- url_for() — CSS/JS ka sahi path deta hai

🎨 CSS — Design + 15+ Animations

style.css — website ko sundar aur jaandar banata hai

Design Techniques

- Glassmorphism — card par blur + transparency
- Neon glow — box-shadow se chamak
- Gradient text — title me 3 rang badalte hain
- Dark theme — #07071a background
- CSS variables — ek jagah rang define, jagah jagah use
- Backdrop filter — background blur effect

Animations List

- @keyframes float1/2/3 — orbs ka float
- @keyframes bgShift — background ka rang
- @keyframes letterIn — title ke letters
- @keyframes caretBlink — typewriter cursor
- @keyframes shine — button par light sweep
- @keyframes scanSweep — AI scan line
- @keyframes glitchIn — hologram result
- @keyframes eqBounce — thinking bars
- @keyframes neonFlicker — credit sign

JavaScript — Website ki Jaan

script.js — sab kuch chalata hai (500+ lines)

1. Background Animation

Canvas par 70 particles move + connect.
Mouse ke paas cyan lines.
Shooting stars aati rehti hain.

2. Title + Typewriter

Har letter span me wrap.
Ek-ek karke uchhal ke aata hai.
Subtitle akshar-akshar likhta hai.

3. 3D Tilt + Cursor

Card mouse ke saath tilt.
Custom neon dot + ring cursor.
Orbs parallax move karti hain.

4. Form + Fetch

8 fields ka data collect.
fetch('/predict') se JSON bhej.
Response me prediction padh.

5. Result Animations

Gauge ring bharti hai (SVG).
% count-up hota hai.
Placed ho to confetti barish.

6. AI Scan + Glitch

Scan line card par sweep.
Fields sequence me highlight.
Result hologram glitch se aata hai.

JavaScript Code — Form Submit

User click karta hai → data jaata hai → result aata hai

```
// script.js – form submit hone par

document.getElementById("predictForm")
.addEventListener("submit", async function (e) {
  e.preventDefault();   // page reload mat ho

  // 1. Saari fields ka data lo
  const data = {
    tenth_percent:  document.getElementById("tenth_percent").value,
    twelfth_percent: document.getElementById("twelfth_percent").value,
    cgpa:           document.getElementById("cgpa").value,
    internships:    document.getElementById("internships").value,
    projects:       document.getElementById("projects").value,
    aptitude_score: document.getElementById("aptitude_score").value,
    communication_score: ...,
    work_experience: ...,
  };

  startScan();  // AI scan animation chalu

  // 2. Flask server ko bhejo
  const response = await fetch("/predict", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data),
  });

  // 3. Result lo aur dikhao
  const result = await response.json();
  gaugeFill.style.strokeDashoffset = ...;  // ring bharo
```

3 Steps Me

- data object — sab fields collect
- fetch() — server ko JSON bhejta hai
- await — response ka wait karta hai
- result.prediction — Placed/Not
- result.probability — % chance
- startScan() — AI scan animation
- countUp() — % smooth badhta hai
- fireConfetti() — celebration!

▶️ Kaise Chalayein aur Deploy Karein

2 command me website apne computer par

Computer Par Chalane Ka

```
# Step 1 — model train (ek baar)
python train_model.py

# Step 2 — website chalao
python app.py
```

Phir browser me kholo: http://127.0.0.1:5000

Dusre device (same WiFi) ke liye:
Terminal me jo IP print hota hai wahi use karo.

Credentials: vikas yadav / pit jaunpur

Free Deploy (Render.com)

```
# Build command
pip install -r requirements.txt

# Start command
gunicorn app:app --bind 0.0.0.0:$PORT

# Instance: Free
```

GitHub → Render → Web Service → repo select → Deploy.
Free plan: 15 min idle par website so jati hai (cold start).

✨ Website ki Khaas Baatein

Ek line me: simple code + tagda ML + zabardast animations

- 🧠 ML Powered — Random Forest (87% accuracy)
- 🔒 Login Security — session + decorator
- 🎯 8 Features se accurate prediction
- 📊 Neon gauge + count-up percentage
- 🎉 Confetti celebration (Placed ho to)
- 🔍 AI scan animation (predict karte waqt)
- 🎨 15+ custom CSS animations
- 🖱️ Custom neon cursor (dot + ring)
- 🫧 Glassmorphism + 3D tilt card
- 🚀 Zero external libraries — pure CSS/JS
- 📱 Mobile friendly + reduced motion support
- 🌐 Deploy ready — Render free hosting

Total Code: app.py (119 lines) + train_model.py (89) + index.html (118) + login.html (271) + style.css (868) + script.js (593)
≈ 2000+ lines — sab handwritten, koi library nahi!

Thank You!

Koi sawaal ho to poochh lo 🙂

Placement Predictor • Created by Vikas Yadav